

Exploratory Data Analysis (EDA) Report

Project Title

Cryptocurrency Volatility Prediction Using Machine Learning

Introduction

Cryptocurrency markets are highly dynamic and volatile. Predicting volatility helps traders and financial institutions manage risks, optimize portfolios, and make informed investment decisions.

This project analyzes historical cryptocurrency market data and develops a machine learning model to predict volatility using features derived from OHLC prices, trading volume, and market capitalization.

Dataset Description

The dataset contains historical daily records for multiple cryptocurrencies.

Feature	Description
Date	Trading date
Symbol	Cryptocurrency ticker
Open	Opening price
High	Highest price
Low	Lowest price
Close	Closing price
Volume	Trading volume
Market Cap	Total market capitalization

Data Preprocessing

Missing Value Handling

- Removed rows with excessive null values

- Forward-filled missing prices
- Replaced missing volume with median values

Data Cleaning

- Converted date column to datetime format
- Removed duplicate entries
- Sorted data chronologically

Feature Scaling

Numerical features were normalized using:

- StandardScaler

Feature Engineering

Several features were created to improve prediction performance.

Volatility Feature

$$Volatility = (High - Low) / Open$$

Additional Engineered Features

7-day Moving Average
14-day Moving Average
Rolling Standard Deviation
Volume-to-MarketCap Ratio
Bollinger Bands

Exploratory Data Analysis

Price Trend Analysis

Bitcoin and Ethereum showed strong long-term upward trends.
Sudden spikes and crashes indicate high volatility periods.

Volume Analysis

Trading volume increased significantly during market rallies.
High volume often correlated with increased volatility.

Correlation Analysis

Key observations:

- High and Close prices are strongly positively correlated.
- Volatility has moderate correlation with trading volume.
- Market capitalization influences price stability.

Distribution Analysis

- Cryptocurrency returns are non-normal and highly skewed.
- Presence of extreme outliers confirms unstable market behavior.

Visualization Summary

- Line plots for price trends
- Heatmaps for correlation analysis

Key Insights

1. Cryptocurrency markets exhibit extreme fluctuations.
2. Trading volume is an important predictor of volatility.
3. Technical indicators improve model learning capability.
4. Short-term moving averages capture rapid market changes effectively.

Conclusion

EDA helped identify:

- Important predictive features
- Market behavior patterns
- Relationships between price and volatility

These insights guided feature engineering and model selection for volatility prediction.